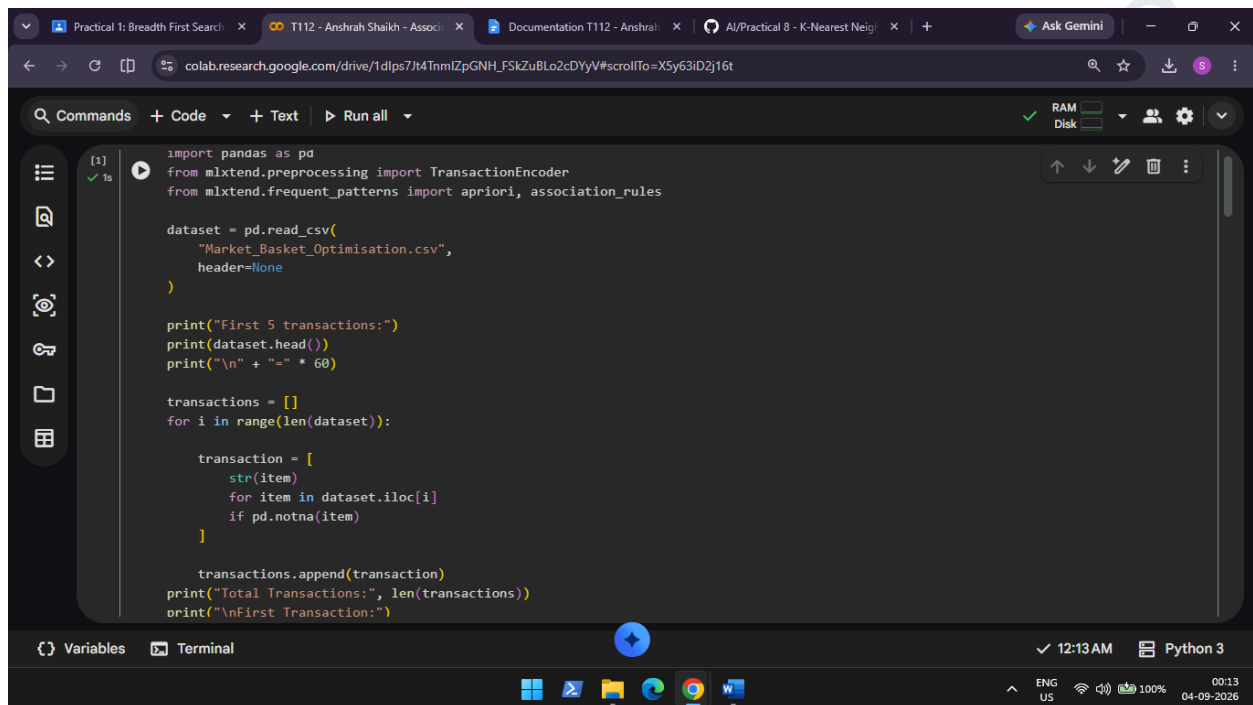


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AIM: Association Rule Mining

- Implement the Association Rule Mining algorithm (e.g., Apriori) to find frequent itemsets.
- Generate association rules from the frequent itemsets and calculate their support and confidence.
- Interpret and analyze the discovered association rules.

INPUT:



```
import pandas as pd
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import apriori, association_rules

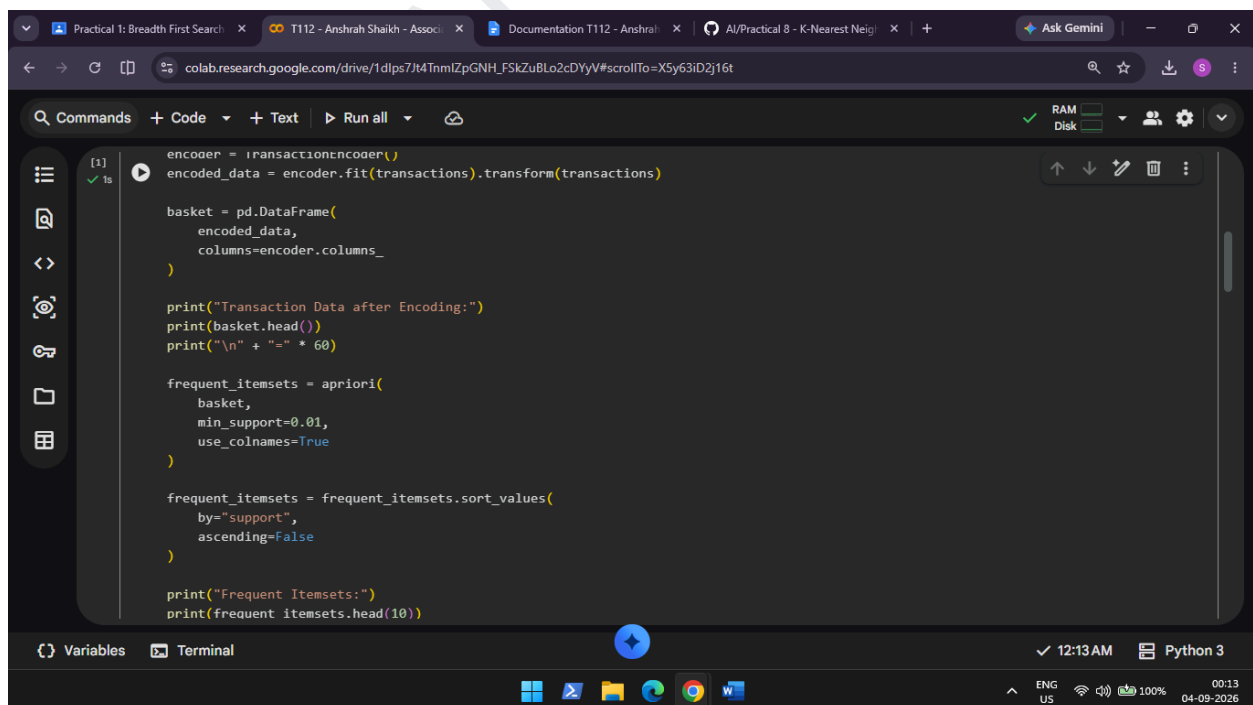
dataset = pd.read_csv(
    "Market_Basket_Optimisation.csv",
    header=None
)

print("First 5 transactions:")
print(dataset.head())
print("\n" + "=" * 60)

transactions = []
for i in range(len(dataset)):

    transaction = [
        str(item)
        for item in dataset.iloc[i]
        if pd.notna(item)
    ]

    transactions.append(transaction)
print("Total Transactions:", len(transactions))
print("\nFirst Transaction:")
```



```
encoder = TransactionEncoder()
encoded_data = encoder.fit(transactions).transform(transactions)

basket = pd.DataFrame(
    encoded_data,
    columns=encoder.columns_
)

print("Transaction Data after Encoding:")
print(basket.head())
print("\n" + "=" * 60)

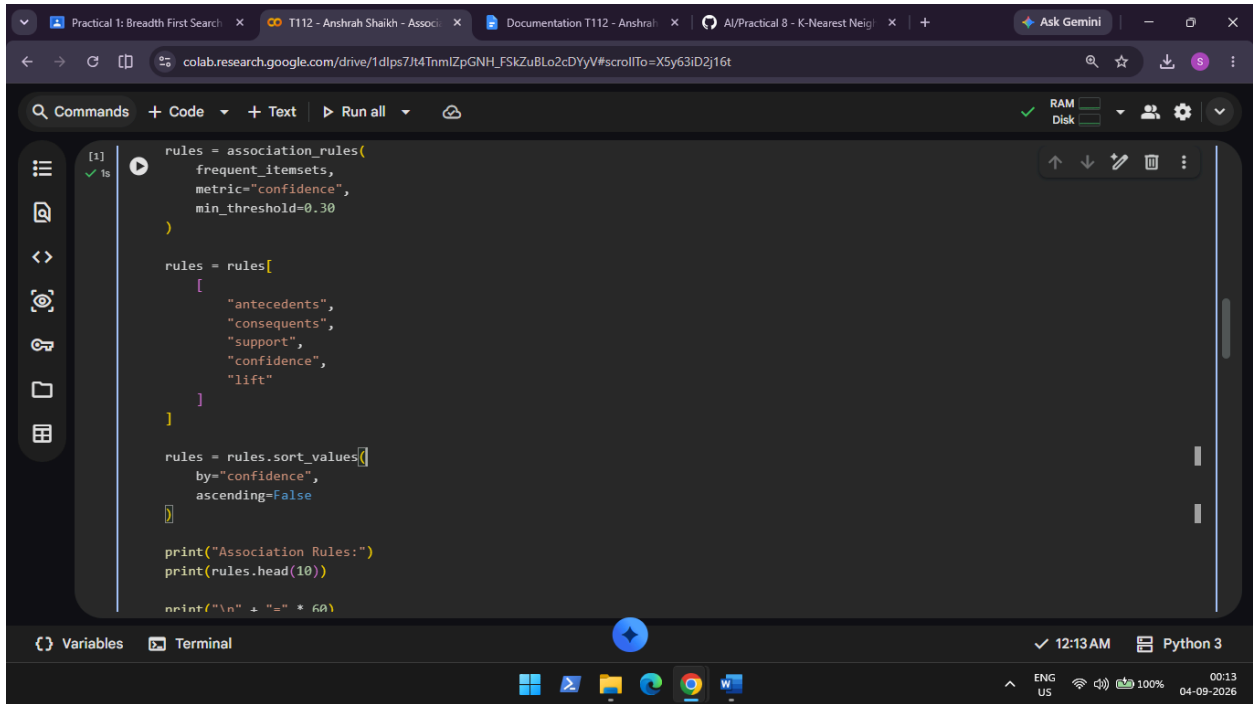
frequent_itemsets = apriori(
    basket,
    min_support=0.01,
    use_colnames=True
)

frequent_itemsets = frequent_itemsets.sort_values(
    by="support",
    ascending=False
)

print("Frequent Itemsets:")
print(frequent_itemsets.head(10))
```

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```
[1] rules = association_rules(frequent_itemsets, metric="confidence", min_threshold=0.30)

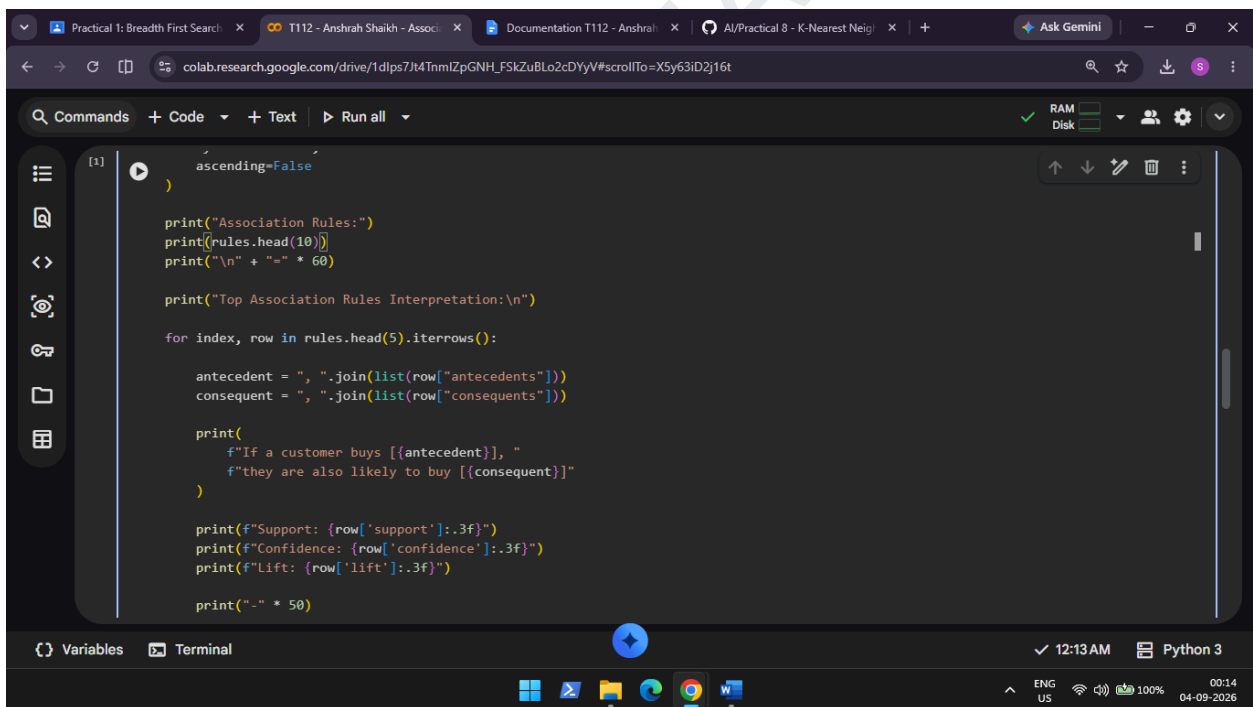
rules = rules[
    [
        "antecedents",
        "consequents",
        "support",
        "confidence",
        "lift"
    ]
]

rules = rules.sort_values(
    by="confidence",
    ascending=False
)

print("Association Rules:")
print(rules.head(10))

print("\n" + "=" * 60)
```

Variables Terminal 12:13 AM Python 3



```
ascending=False
)

print("Association Rules:")
print(rules.head(10))
print("\n" + "=" * 60)

print("Top Association Rules Interpretation:\n")

for index, row in rules.head(5).iterrows():

    antecedent = ", ".join(list(row["antecedents"]))
    consequent = ", ".join(list(row["consequents"]))

    print(
        f"If a customer buys [{antecedent}], "
        f"they are also likely to buy [{consequent}]"
    )

    print(f"Support: {row['support']:.3f}")
    print(f"Confidence: {row['confidence']:.3f}")
    print(f"Lift: {row['lift']:.3f}")

    print("-" * 50)
```

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OUTPUT:

```
colab.research.google.com/drive/1dlps7Jt4TnmIZpGNH_FSkZuBLo2cDYyV#scrollTo=XSy63iD2y16t

Commands + Code + Text Run all RAM Disk

First 5 transactions:
0 0 1 2 3 4 \
0 shrimp almonds avocado vegetables mix green grapes
1 burgers meatballs eggs NaN NaN NaN
2 chutney NaN NaN NaN NaN NaN
3 turkey avocado NaN NaN NaN NaN
4 mineral water milk energy bar whole wheat rice green tea

0 5 6 7 8 9 \
0 whole wheat flour yams cottage cheese energy drink tomato juice
1 NaN NaN NaN NaN NaN NaN
2 NaN NaN NaN NaN NaN NaN
3 NaN NaN NaN NaN NaN NaN
4 NaN NaN NaN NaN NaN NaN

0 10 11 12 13 14 15 \
0 low fat yogurt green tea honey salad mineral water salmon
1 NaN NaN NaN NaN NaN NaN
2 NaN NaN NaN NaN NaN NaN
3 NaN NaN NaN NaN NaN NaN
4 NaN NaN NaN NaN NaN NaN

0 16 17 18 19
0 antioxidant juice frozen smoothie spinach olive oil
1 NaN NaN NaN NaN
2 NaN NaN NaN NaN
3 NaN NaN NaN NaN
```

```
colab.research.google.com/drive/1dlps7Jt4TnmIZpGNH_FSkZuBLo2cDYyV#scrollTo=XSy63iD2y16t

Commands + Code + Text Run all RAM Disk

Total Transactions: 7501
First Transaction:
['shrimp', 'almonds', 'avocado', 'vegetables mix', 'green grapes', 'whole wheat flour', 'yams', 'cottage cheese', 'energy drink', 'tomato juice']

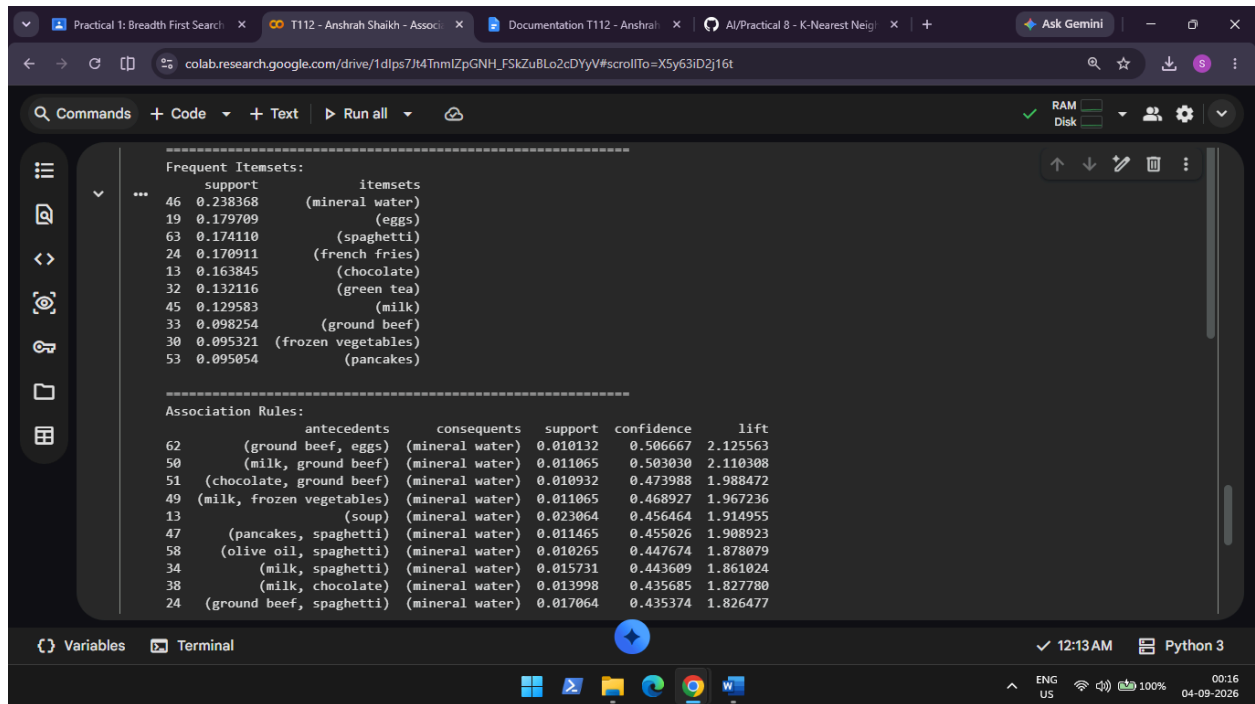
Transaction Data after Encoding:
0 asparagus almonds antioxidant juice asparagus avocado babies food \
0 False True False True False True False
1 False False False False False False False
2 False False False False False False False
3 False False False False False True False
4 False False False False False False False

0 bacon barbecue sauce black tea blueberries ... turkey vegetables mix \
0 False False False False ... False True
1 False False False False ... False False
2 False False False False ... False False
3 False False False False ... True False
4 False False False False ... False False

0 water spray white wine whole wheat flour whole wheat pasta \
0 False False False True False
1 False False False False False
2 False False False False False
3 False False False False False
4 False False False False False
```

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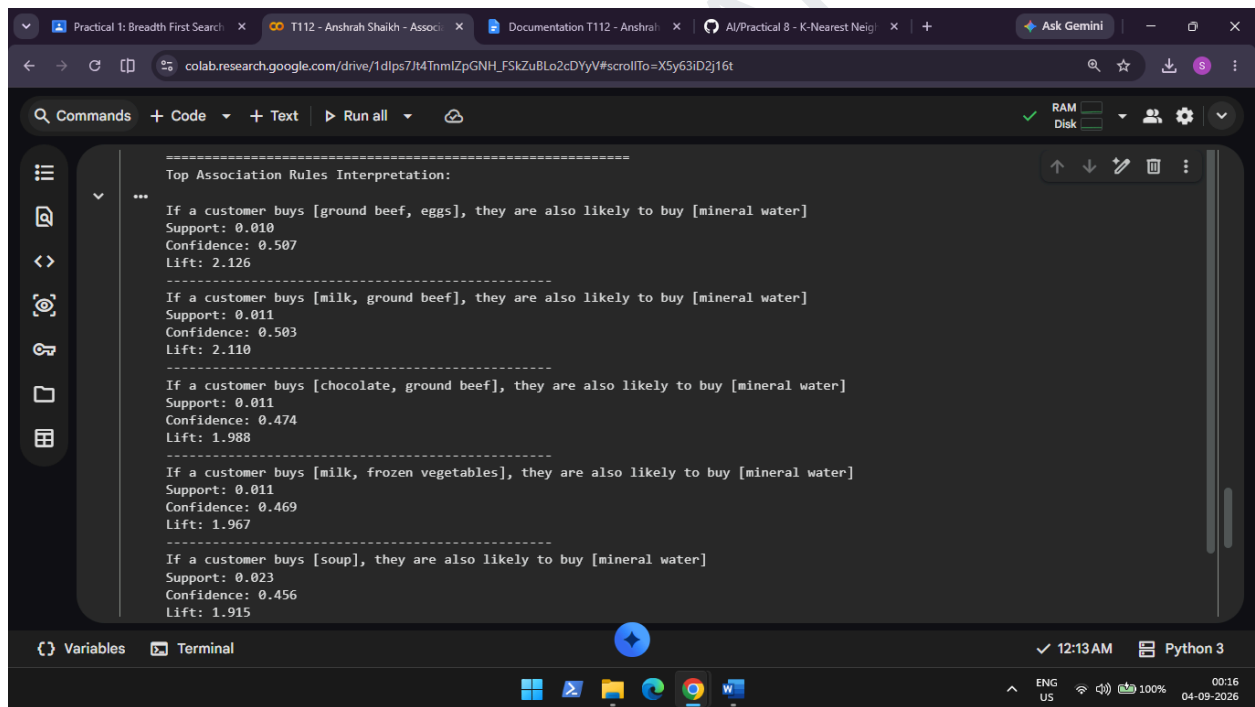


The screenshot shows a Google Colab notebook interface. The top bar includes tabs for 'Practical 1: Breadth First Search', 'T112 - Anshrah Shaikh - Associ...', 'Documentation T112 - Anshrah...', and 'AI/Practical 8 - K-Nearest Neigh...'. The address bar shows the Colab URL. The notebook content is displayed in a dark-themed editor with a left sidebar containing icons for file management and search. The code is as follows:

```
=====
Frequent Itemsets:
support      itemsets
46 0.238368 (mineral water)
19 0.179709 (eggs)
63 0.174110 (spaghetti)
24 0.170911 (french fries)
13 0.163845 (chocolate)
32 0.132116 (green tea)
45 0.129583 (milk)
33 0.098254 (ground beef)
30 0.095321 (frozen vegetables)
53 0.095054 (pancakes)
=====

Association Rules:
antecedents consequents support confidence lift
62 (ground beef, eggs) (mineral water) 0.010132 0.506667 2.125563
50 (milk, ground beef) (mineral water) 0.011065 0.503030 2.110308
51 (chocolate, ground beef) (mineral water) 0.010932 0.473988 1.988472
49 (milk, frozen vegetables) (mineral water) 0.011065 0.468927 1.967236
13 (soup) (mineral water) 0.023064 0.456464 1.914955
47 (pancakes, spaghetti) (mineral water) 0.011465 0.455026 1.908923
58 (olive oil, spaghetti) (mineral water) 0.010265 0.447674 1.878079
34 (milk, spaghetti) (mineral water) 0.015731 0.443609 1.861024
38 (milk, chocolate) (mineral water) 0.013998 0.435685 1.827780
24 (ground beef, spaghetti) (mineral water) 0.017064 0.435374 1.826477
```

The bottom status bar shows 'Variables', 'Terminal', '12:13 AM', and 'Python 3'.



The screenshot shows the same Google Colab notebook interface. The code is as follows:

```
=====
Top Association Rules Interpretation:

If a customer buys [ground beef, eggs], they are also likely to buy [mineral water]
Support: 0.010
Confidence: 0.507
Lift: 2.126
-----

If a customer buys [milk, ground beef], they are also likely to buy [mineral water]
Support: 0.011
Confidence: 0.503
Lift: 2.110
-----

If a customer buys [chocolate, ground beef], they are also likely to buy [mineral water]
Support: 0.011
Confidence: 0.474
Lift: 1.988
-----

If a customer buys [milk, frozen vegetables], they are also likely to buy [mineral water]
Support: 0.011
Confidence: 0.469
Lift: 1.967
-----

If a customer buys [soup], they are also likely to buy [mineral water]
Support: 0.023
Confidence: 0.456
Lift: 1.915
```

The bottom status bar shows 'Variables', 'Terminal', '12:13 AM', and 'Python 3'.

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